

Step-by-Step Setup for Virtual SOC Lab

✓ 1. Install VirtualBox (or VMware Workstation Player)

- Download: <https://www.virtualbox.org/>
 - Install VirtualBox + Extension Pack
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✓ 2. Download OS ISO Files

VM	Download Link
Windows 10 ISO	https://www.microsoft.com/software-download/windows10 (choose "Create installation media")
Kali Linux	https://www.kali.org/get-kali/#kali-virtual-machines (use ISO or prebuilt VBox image)
Ubuntu Server	https://ubuntu.com/download/server (LTS version recommended)

✓ 3. Create Virtual Machines in VirtualBox

Windows 10 (Victim)

- **Name:** Win10-Victim
- **OS Type:** Windows 10 (64-bit)
- **RAM:** 4096 MB (minimum)
- **Disk:** 60 GB (dynamically allocated)
- **Network:** *Internal Network* (e.g. name it LabNet)
- Attach the ISO and install Windows.
- Optional: Disable auto-updates to keep VM stable.

Kali Linux (Attacker)

- **Name:** Kali-Attacker
- **OS Type:** Debian / Linux (64-bit)
- **RAM:** 4096–8192 MB
- **Disk:** 40 GB
- **Network:** *Internal Network* (LabNet)
- Install Kali or import prebuilt image.
- Run `sudo apt update && sudo apt full-upgrade`

🖥️ Ubuntu Server (SIEM Host)

- **Name:** Ubuntu-SIEM
 - **OS Type:** Ubuntu (64-bit)
 - **RAM:** 4096–8192 MB
 - **Disk:** 40–60 GB
 - **Network:** *Internal Network* (LabNet)
 - Install Ubuntu Server with OpenSSH server.
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✅ 4. Enable Network Communication Between VMs

- In **VirtualBox**, all three VMs should be attached to the same **Internal Network** (e.g., LabNet)
- This isolates them from the internet but allows communication **between** VMs.

Test connectivity once all are running:

```
bash
CopyEdit
# From Kali
ping <Ubuntu IP>
ping <Windows IP>

# From Windows
ping <Kali IP>
```

If ping fails:

- On Linux: Check `ip a` for IP assignment and use `sudo ufw disable` temporarily.
 - On Windows: Allow ICMP Echo Request in Windows Firewall.
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✅ 5. Take Snapshots

Before installing SIEM or launching attacks, take clean snapshots of each VM. This lets you roll back if needed.

✅ 6. Start Installing Tools Per Role

- **Ubuntu Server** → Install Wazuh/ELK/Splunk + Filebeat
- **Windows** → Install Winlogbeat
- **Kali** → Install attack tools (Nmap, Hydra, Metasploit)

Optional: Shared Folder Setup

You can enable shared folders in VirtualBox to transfer files between host and VM (e.g., agent configs, scripts).